

CHETANA MURALIDHARAN

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EDUCATION

San Jose State University <i>M.S., Applied Data Intelligence GPA: 3.88/4.00</i>	San Jose, CA August 2024 – May 2026
Visvesvaraya Technological University <i>B.E., Computer Science & Engineering CGPA: 8.5/10.00</i>	KA, India August 2017 – August 2021

EXPERIENCE

Instructional Student Assistant(ISA) - DATA 266: Generative Models <i>San Jose State University</i>	January 2026 – May 2026 <i>San Jose, CA</i>
- Supported 35+ graduate students by diagnosing and resolving implementation failures across transformer, RAG, diffusion, and agentic workflow assignments, improving pipeline correctness and reducing repeated submission errors across the cohort. - Accelerated assignment quality across 100+ student submissions by delivering technical feedback on LLM fine-tuning, vector db pipelines, RAG, and multi-agent system design, cutting failure patterns and improving readiness of student implementations.	
Data Engineer (ETL Developer) <i>Accenture</i>	August 2021 – July 2024 <i>India</i>
- Reduced manual data preparation effort by 70% by designing and automating 60+ production ETL/ELT pipelines in Python and SQL on Snowflake and AWS S3, delivering analytics-ready OLAP datasets to business and BI teams at scale. - Cut data refresh latency by 85% (from 6 hours to 45 minutes) by migrating full-load pipelines to CDC-based incremental ingestion and optimizing SQL transformations across APAC regional data sources, enabling near real-time decision-making on globally distributed datasets for cross-regional business stakeholders. - Increased data accuracy to 96-98% across all pipelines by implementing automated validation checks, schema enforcement, and reconciliation frameworks, establishing data quality and observability standards for downstream consumption. - Reduced ad-hoc data requests by 60% by designing star schema dimensional models with a reusable semantic layer of core business metrics, enabling self-service analytics and eliminating inconsistent KPI definitions across BI teams.	

PROJECTS

LineageIQ: dbt Observability and AI Blast Radius Platform - Enforced data quality across 9 dbt models by implementing 15+ tests covering uniqueness, referential integrity, null checks, and accepted values, preventing bad data from propagating to downstream dashboards and analytical consumers. - Reduced downstream blast radius triage from mins to secs by feeding Airflow-orchestrated schema drift detections, NetworkX-computed lineage depth, quality metrics into an LLM, producing a CRITICAL-to-LOW ranked impact report per schema change. - Automated schema drift detection across a Snowflake schema registry by baseline-comparing column definitions after every Airflow-orchestrated dbt build (production DAG scheduling with dependency management and failure alerting), flagging HIGH-severity column drops and type changes before pipeline failures could occur.	April 2026 – June 2026
WeaveNet: Communication Analytics Platform - Reduced dataset noise by 56% and retained 226K high-quality records from 517K+ enterprise emails by building modular parsing, deduplication, topic enrichment, and thread reconstruction pipelines, producing a clean analytical foundation for graph modeling. - Designed a heterogeneous graph of 764K+ nodes and 2.86M relationships by modeling communication data into structured node and edge schemas in Python, creating a reusable analytical layer for collaboration, topic, and communication pattern analysis. - Reduced graph query latency to under 100ms across 40+ traversal and analytical queries by implementing indexed data structures and optimized retrieval workflows, improving time-to-insight for exploratory analytics at scale.	August 2025 – May 2026
Smoke & Smog: Fire Incidents & Air Quality Analytics Pipeline - Built an end-to-end analytics engineering platform processing 4M+ environmental records from NASA, weather, and AQ sources using Airflow orchestration, dbt transformations, and Snowflake, producing validated OLAP-ready models across 25 U.S. cities. - Strengthened data reliability across 7 dbt transformation models by implementing 15+ quality tests and freshness checks, enforcing a semantic layer of validated, reusable analytical assets aligned with DataOps best practices. - Delivered two Power BI dashboards with DAX measures and a composite Environmental Distress Score derived from 6 lag-based AQI metrics, enabling faster identification of high-risk regions and delayed pollution impact patterns.	February 2025 – May 2025

TECHNICAL SKILLS

Data Engineering: ETL/ELT Pipelines, CDC Pipelines, Incremental Processing, Real-Time Pipelines, Streaming Ingestion, Apache Airflow, dbt (Models, Tests, Snapshots, Freshness Checks), Apache Spark, Kafka, Orchestration Frameworks
Analytics Engineering: Semantic Layer Design, Dimensional Modeling (Star/Snowflake Schema), Data Quality Testing, Schema Drift Detection, Pipeline Observability, DataOps, CI/CD for Data Workflows, OLTP/OLAP Processing, Self-Service Analytics, KPI Development, Data Governance
Data Warehousing and Cloud: Snowflake (SnowPro Core Certified), AWS S3, BigQuery, Databricks, Data Lakes, MySQL, Postgres
BI & Visualization: Power BI, DAX, Tableau, Dashboard Development, KPI Reporting
AI & LLM Tools: Prompt Engineering, RAG, LangChain, Agentic Workflows, Vector Databases, GraphRAG, NetworkX
Languages & Dev Tools: Python, SQL, Java, Git, CI/CD, Agile/Scrum
Compliance and Access Controls: PHI Data Handling, HIPAA-Aligned Access Controls, Schema Enforcement, Data Reconciliation